spirit to the setup at Ronnie the Bear's casino from Chapter 7. Rather than pairs of roulette balls, the CHSH experiment uses pairs of photons with entangled polarizations. The CHSH experiment proposed sending each photon through a polarizer pointing in one of two different directions (see Figures 9.1 and 9.2), and repeating the experiment with many pairs of entangled photons. Just as each roulette ball at the casino landed on red or black, each photon would either pass through a polarizer or be blocked by it. Comparing the behavior of many pairs of these photons would test Bell's theorem. If each pair of entangled photons had a prearranged plan for how to behave at each of the two polarizers, then the results would satisfy Bell's inequality. But quantum physics predicted that the photons would violate Bell's inequality, just as the roulette balls did at Ronnie's casino.

Whatever the outcome, Clauser, Shimony, and the others knew that the experiment would be immensely important: either it would show that quantum physics was wrong, blowing up a cornerstone of modern physics and garnering a nearly instant Nobel Prize, or the quantum predictions would be correct and Bell's inequality would be violated, meaning nature must be nonlocal (or something even stranger is going on). Clauser was still optimistic that the experiment would show that Bell's inequality was not violated—he estimated a 50 percent chance that quantum physics was wrong. But, like Bell, Shimony suspected that the experimental results would match the quantum-mechanical prediction—as did nearly everyone else. "Aharonov bet [Clauser] $100 against $1 that the outcome would favor quantum mechanics," Shimony wrote to Wigner. "I am much more conservative than Clauser in estimating the outcome. However, in view of the difficulty of the measurement problem in quantum mechanics and of the fact that hidden variable theories do offer a solution, I would not entirely discount the possibility of an outcome in favor of [local] hidden variables."

